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Overall Survival with Pembrolizumab in Early-Stage Triple-Negative Breast Cancer

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ABSTRACT

BACKGROUND
In patients with early-stage triple-negative breast cancer, the phase 3 KEYNOTE-522 trial showed significant improvements in pathological complete response and event-free survival with the addition of pembrolizumab to platinum-containing chemotherapy. Here we report the final results for overall survival.

METHODS
We randomly assigned, in a 2:1 ratio, patients with previously untreated stage II or III triple-negative breast cancer to receive neoadjuvant therapy with four cycles of pembrolizumab (at a dose of 200 mg) or placebo every 3 weeks plus paclitaxel and carboplatin, followed by four cycles of pembrolizumab or placebo plus doxorubicin–cyclophosphamide or epirubicin–cyclophosphamide. After definitive surgery, patients received adjuvant pembrolizumab (pembrolizumab–chemotherapy group) or placebo (placebo–chemotherapy group) every 3 weeks for up to nine cycles. The primary end points were pathological complete response and event-free survival. Overall survival was a secondary end point.

RESULTS
Of the 1174 patients who underwent randomization, 784 were assigned to the pembrolizumab–chemotherapy group and 390 to the placebo–chemotherapy group. At the data-cutoff date (March 22, 2024), the median follow-up was 75.1 months (range, 65.9 to 84.0). The estimated overall survival at 60 months was 86.6% (95% confidence interval [CI], 84.0 to 88.8) in the pembrolizumab–chemotherapy group, as compared with 81.7% (95% CI, 77.5 to 85.2) in the placebo–chemotherapy group ($P=0.002$). Adverse events were consistent with the established safety profiles of pembrolizumab and chemotherapy.

CONCLUSIONS
Neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant pembrolizumab resulted in a significant improvement, as compared with neoadjuvant chemotherapy alone, in overall survival among patients with early-stage triple-negative breast cancer. (Funded by Merck Sharp and Dohme, a subsidiary of Merck [Rahway, NJ]; KEYNOTE-522 ClinicalTrials.gov number, NCT03036488.)

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*The complete list of principal investigators in the KEYNOTE-522 trial is provided in the Supplementary Appendix, available at NEJM.org.

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CME

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Breast cancer is the most common neoplasm affecting women and a leading cause of death.[1,2] Several molecular subtypes of breast cancer are recognized, broadly categorized by the expression of cell-surface receptors. Triple-negative breast cancer lacks expression of estrogen receptor, progesterone receptor, and human epidermal growth factor receptor 2.[3] Owing to the absence of therapeutic targets, treatment of triple-negative breast cancer has traditionally been challenging.[4] In addition, triple-negative breast cancer is more aggressive than other breast cancer subtypes[5,6] and is associated with an earlier onset and an increased risk of early (primarily distant) recurrences and shorter overall survival despite the use of curative-intent systemic chemotherapy.[7,8]

The programmed death 1 (PD-1) inhibitor pembrolizumab is approved for the treatment of early-stage triple-negative breast cancer on the basis of results from the phase 3 KEYNOTE-522 trial of neoadjuvant and adjuvant pembrolizumab treatment.[9] At the first interim analysis, the addition of pembrolizumab to platinum-containing neoadjuvant chemotherapy resulted in a significant increase in pathological complete response, defined as pathological stage ypT0–Tis ypN0 (indicating no residual invasive cancer in the complete resected breast specimen and all sampled regional lymph nodes) at the time of definitive surgery, of 13.6 percentage points ($P<0.001$).[10] At the fourth interim analysis after a median follow-up of 39.1 months, the addition of pembrolizumab to neoadjuvant chemotherapy followed by adjuvant pembrolizumab resulted in a significant improvement in event-free survival (hazard ratio for event or death, 0.63; $P<0.001$).[11] On the basis of these results, treatment guidelines recommend the use of neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant pembrolizumab for high-risk, early-stage triple-negative breast cancer.[12-14] In this article, we report the final overall survival results from the KEYNOTE-522 trial.

Methods

Patients

The full methods were described previously.[10,11] In brief, patient eligibility criteria included centrally confirmed triple-negative breast cancer as defined by the American Society of Clinical Oncology–College of American Pathologists guidelines[15-17]; newly diagnosed, previously untreated, nonmetastatic disease, which was defined as combined primary tumor (T) and regional lymph node (N) involvement, according to the American Joint Committee on Cancer staging criteria (7th edition),[18] as assessed by the trial investigator on the basis of radiologic or clinical assessment (T1c N1–2 or T2–4 N0–2 disease; see below); an Eastern Cooperative Oncology Group performance-status score[19] of 0 or 1 (range, 0 to 5, with higher scores indicating greater disability); and adequate organ function. Complete eligibility criteria are listed in the trial protocol, which is available with the full text of this article at NEJM.org.

Trial Design and Treatment

KEYNOTE-522 is a phase 3, double-blind, randomized, placebo-controlled trial being conducted at 181 sites (plus 2 satellite sites) in 21 countries throughout Europe, North America, Asia, and Latin America. Patients were stratified before randomization on the basis of nodal status (positive or negative), tumor size (T1 [diameter, >1.0 cm to 2.0 cm] to T2 [diameter, >2.0 cm to 5.0 cm] or T3 [diameter, >5.0 cm] to T4 [locally advanced disease]), and frequency of carboplatin administration (once weekly or once every 3 weeks). Patients were randomly assigned, in a 2:1 ratio, to neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant pembrolizumab (pembrolizumab–chemotherapy group) or neoadjuvant placebo plus chemotherapy followed by adjuvant placebo (placebo–chemotherapy group). Randomization was performed with the use of a central interactive voice-response system with an integrated Web-response system.

Patients were treated in a neoadjuvant phase and an adjuvant phase. In the neoadjuvant phase, patients received four cycles of an intravenous infusion of pembrolizumab (at a dose of 200 mg) or placebo once every 3 weeks; all patients also received paclitaxel (80 mg per square meter of body-surface area once weekly) and carboplatin (at a dose based on an area under the concentration–time curve of 5 mg per milliliter per minute once every 3 weeks, or 1.5 mg per milliliter per minute, administered once weekly) for the first 12 weeks (i.e., the first neoadjuvant treatment). Patients then received four cycles of pembrolizumab or placebo; all patients also received doxorubicin (60 mg per square meter) or epirubicin (90 mg per square meter) and cyclophosphamide (600 mg per square meter), administered once every 3 weeks for the subsequent 12 weeks (i.e., the second neoadjuvant treatment).

Patients underwent definitive surgery (breast conservation or mastectomy with sentinel lymph-node evaluation or axillary dissection) at 3 to 6 weeks after the last treatment cycle of the neoadjuvant phase. In the adjuvant phase, patients received radiation therapy as indicated and either pembrolizumab or placebo, administered once every 3 weeks for up to nine cycles. Adjuvant therapy with capecitabine was not allowed. Trial treatment was discontinued for disease progression that precluded definitive surgery, disease recurrence, or unacceptable toxic effects.

ASSESSMENTS

Pathological complete response was assessed after patients completed neoadjuvant therapy.[10] Event-free survival was determined by an investigator who was unaware of the trial-group assignments.[11] Follow-up for disease status and survival occurred every 3 months for the first 2 years after randomization, every 6 months for years 3 through 5, and annually thereafter.

PD-L1 expression was assessed in formalin-fixed tumor samples at a central laboratory with the use of the PD-L1 IHC 22C3 pharmDx assay (Agilent) and characterized by the combined positive score, which was defined as the number of PD-L1–positive cells (tumor cells, lymphocytes, and macrophages) divided by the total number of tumor cells, multiplied by 100. Patients were eligible for the trial regardless of PD-L1 expression status.

Adverse events were monitored throughout the trial and for 30 days after treatment discontinuation, with serious adverse events being monitored for 90 days after treatment discontinuation. Adverse events were graded according to the Common Terminology Criteria for Adverse Events, version 4.0, of the National Cancer Institute.[20] Immune-mediated adverse events were determined on the basis of a prespecified list of *Medical Dictionary for Regulatory Activities* (MedDRA) terms.[21]

END POINTS

The two primary end points were pathological complete response, defined as pathological stage ypT0–Tis ypN0 at the time of definitive surgery, and event-free survival, defined as the time from randomization to the first occurrence of disease progression that precluded definitive surgery, local or distant recurrence, a second primary cancer, or death from any cause; results for these end points were reported previously.[10,11] Overall survival, defined as the time from randomization to death from any cause, was a key secondary end point. Safety was evaluated in all the patients who received at least one trial drug or underwent surgery.

TRIAL OVERSIGHT

This trial was developed by a scientific advisory committee and representatives of the sponsor (Merck Sharp and Dohme, a subsidiary of Merck [in Rahway, New Jersey]). An external, independent data and safety monitoring committee provided trial oversight, periodically assessed safety, and assessed efficacy at prespecified interim analyses. The trial protocol and all amendments were approved by the appropriate ethics committee at each participating institution. All the patients provided written informed consent before enrollment.

The authors attest that the trial was conducted in accordance with the protocol, its amendments, and Good Clinical Practice standards. All the authors had access to the data that were used to prepare the manuscript and participated in the writing or critical review and editing of the manuscript. The first draft of the manuscript was written by the first author, with editorial assistance provided by a medical writer employed by the trial sponsor. All the authors approved the manuscript for submission for publication and confirm the accuracy and completeness of the data reported.

STATISTICAL ANALYSIS

The statistical analysis plan and the results for the primary end points of pathological complete response and event-free survival were published previously.[10,11] The present results are from the seventh prespecified interim analysis (data-cutoff date, March 22, 2024), which was calendar-driven and was planned to occur approximately 84 months after the first patient had undergone randomization.

Efficacy analyses were performed in the intention-to-treat population, which included all the patients who underwent randomization. The

Kaplan–Meier method was used to estimate overall survival and event-free survival. Data from patients without documented death at the time of data analysis were censored for overall survival at the date of the last follow-up. Data from patients who did not have an event at the time of data analysis were censored for event-free survival at the date on which they were last known to be alive and event-free. Treatment differences in survival were assessed by means of the stratified log-rank test. Hazard ratios and associated 95% confidence intervals were determined with the use of a stratified Cox proportional-hazards model with Efron's method of tie handling. The 95% confidence intervals associated with between-group differences were not adjusted for multiple comparisons and therefore cannot be used to infer treatment effects. The same stratification factors that were used for randomization were used in all the stratified analyses. For the primary analysis, if the proportional-hazards assumption was not valid, the restricted mean survival time method[22] was performed as a sensitivity analysis and the piecewise unstratified hazard ratios with weighted average hazard ratio were provided.

The graphical method of Maurer and Bretz,[23] as described previously,[10,11] was used to strictly control the type I error rate at a one-sided alpha level of 0.025 across multiple hypotheses and the interim and final analyses (see the Supplementary Appendix, available at NEJM.org). The Lan–DeMets O'Brien–Fleming spending function was used to control the type I error in all the analyses.[24] Overall survival and event-free survival were assessed at the time of this analysis. On the basis of the prespecified multiplicity adjustment strategy, the alpha was passed to overall survival by a step-down approach from event-free survival after both primary end points of event-free survival and pathological complete response had reached statistical significance. At the time of this analysis, overall survival was analyzed with the use of alpha-adjusted hypothesis testing to compare with a prespecified critical value (0.00503). The present results for event-free survival are descriptive only, given that a significant improvement in the pembrolizumab–chemotherapy group as compared with the placebo–chemotherapy group was previously shown at the fourth interim analysis. The trial had 82% power to detect a hazard ratio for death from any cause of 0.70 at a one-sided alpha level of 0.025.

Safety was evaluated in the as-treated population, which included all the patients who had undergone randomization and received at least one trial drug, underwent surgery, or both. The full statistical analysis plan is available in Section 8 of the trial protocol.

Results

Patients and Treatment

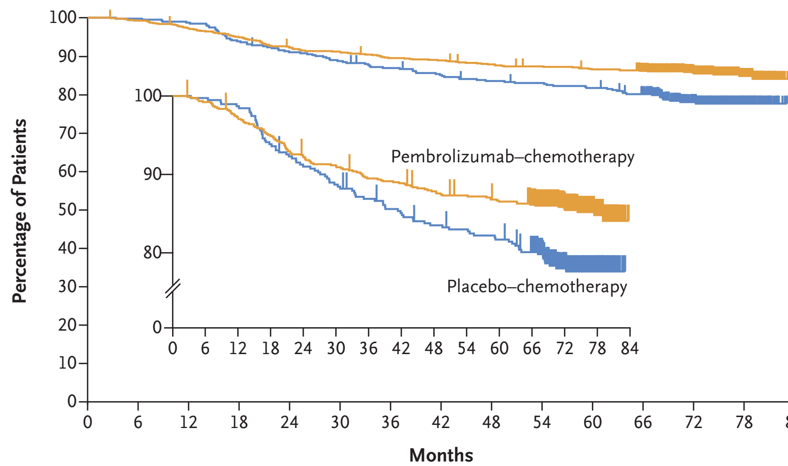
A total of 1174 patients were randomly assigned to receive trial treatment between March 2017 and September 2018, with 784 in the pembrolizumab–chemotherapy group and 390 in the placebo–chemotherapy group (the intention-to-treat population). Of these, 783 patients in the pembrolizumab–chemotherapy group and 389 in the placebo–chemotherapy group received at least one trial drug, underwent surgery, or both (the as-treated population); no patients were still receiving treatment at the time of this analysis (Fig. S1 in the Supplementary Appendix). As reported previously,[10,11] the characteristics of the patients at baseline were generally well balanced between the treatment groups (Table S1). The representativeness of the patient population is shown in Table S2.

At this analysis (data-cutoff date, March 22, 2024), the median duration of follow-up was 75.1 months (range, 65.9 to 84.0). The median duration of treatment exposure and the median number of chemotherapy doses administered were similar in the two treatment groups (Table S3).

Efficacy

As of the data-cutoff date, 200 patients had died (information fraction, 67.3%), including 115 patients (14.7%) in the pembrolizumab–chemotherapy group and 85 patients (21.8%) in the placebo–chemotherapy group ($P=0.002$) (Fig. 1A). According to the prespecified statistical criterion of an alpha level of 0.00503, a significant improvement in overall survival was observed in the pembrolizumab–chemotherapy group as compared with the placebo–chemotherapy group; thus, no formal testing of overall survival will be performed at a later analysis. The estimated overall survival in the pembrolizumab–chemotherapy group and the placebo–chemotherapy group,

A Overall Survival According to Treatment Group in the Intention-to-Treat Population



No. at Risk

	0	6	12	18	24	30	36	42	48	54	60	66	72	78	84
Pembrolizumab–chemotherapy	784	777	760	742	720	712	698	693	683	677	670	656	448	176	0
Placebo–chemotherapy	390	389	385	366	354	345	336	328	321	318	313	300	199	82	0

B Subgroup Analyses of Overall Survival

Subgroup	Pembrolizumab–Chemotherapy	Placebo–Chemotherapy	Difference in 5-Year Overall Survival (95% CI)
	no. of patients who died/total no. (%)		*percentage points*
Overall	115/784 (14.7)	85/390 (21.8)	4.9 (0.3 to 9.4)
Nodal status			
Positive	78/408 (19.1)	56/196 (28.6)	7.2 (0.0 to 14.3)
Negative	37/376 (9.8)	29/194 (14.9)	2.9 (−2.5 to 8.2)
Tumor size			
T1 to T2	54/580 (9.3)	51/290 (17.6)	5.1 (0.5 to 9.6)
T3 to T4	61/204 (29.9)	34/100 (34.0)	4.5 (−6.5 to 15.5)
Carboplatin schedule			
Every 3 wk	46/334 (13.8)	36/167 (21.6)	4.7 (−2.1 to 11.6)
Weekly	68/444 (15.3)	49/220 (22.3)	5.2 (−0.8 to 11.3)
PD-L1 status			
CPS ≥1	92/656 (14.0)	62/317 (19.6)	3.5 (−1.4 to 8.3)
CPS <1	23/128 (18.0)	23/69 (33.3)	11.9 (−0.4 to 24.2)
Age			
<65 yr	93/700 (13.3)	72/342 (21.1)	5.0 (0.3 to 9.7)
≥65 yr	22/84 (26.2)	13/48 (27.1)	2.4 (−12.8 to 17.6)

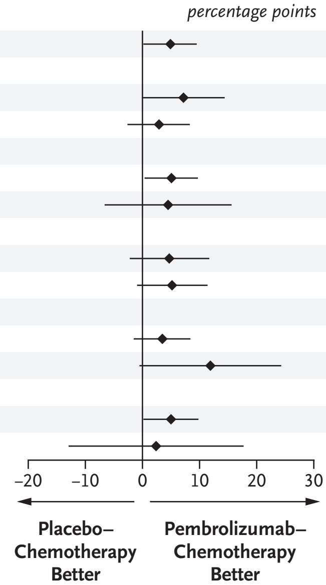


Figure 1. Overall Survival.

Panel A shows Kaplan–Meier estimates of overall survival according to treatment group in the intention-to-treat population. Tick marks represent data censored at the last time that the patient was known to be alive, among patients without documented death. The inset shows the same data on an enlarged y axis. Panel B shows subgroup analyses of overall survival. Kaplan–Meier estimates of 5-year overall survival in prespecified subgroups are shown. The difference in 5-year overall survival and 95% confidence intervals for the difference are reported, under the assumption that Kaplan–Meier estimates of 5-year overall survival in both treatment groups follow the asymptotic normal distribution independently. For tumor size, T1 indicates a diameter of more than 1.0 cm to 2.0 cm, T2 a diameter of more than 2.0 cm to 5.0 cm, T3 a diameter of more than 5.0 cm, and T4 locally advanced disease. The programmed death ligand 1 (PD-L1) combined positive score (CPS) was defined as the number of PD-L1–positive cells (tumor cells, lymphocytes, and macrophages) divided by the total number of tumor cells, multiplied by 100.

respectively, was 89.5% (95% confidence interval [CI], 87.1 to 91.5) and 86.9% (95% CI, 83.1 to 89.9) at 3 years, 87.8% (95% CI, 85.3 to 89.9) and 83.5% (95% CI, 79.4 to 86.9) at 4 years, and 86.6% (95% CI, 84.0 to 88.8) and 81.7% (95% CI, 77.5 to 85.2) at 5 years. The restricted mean survival time in the pembrolizumab–chemotherapy group and the placebo–chemotherapy group, respectively, was 34.2 months and 34.0 months at 3 years (difference, 0.2 months; 95% CI, −0.5 to 0.9), 44.8 months and 44.2 months at 4 years (difference, 0.6 months; 95% CI, −0.5 to 1.8), and 55.3 months and 54.1 months at 5 years (difference, 1.2 months; 95% CI, −0.5 to 2.8). The unstratified piecewise hazard ratio for death was 0.87 (95% CI, 0.57 to 1.32) before the 2-year follow-up and 0.51 (95% CI, 0.35 to 0.75) afterwards. The weighted average hazard ratio with weights of number of events before and after 2-year follow-up was 0.66. Overall survival in the prespecified subgroups is shown in Figure 1B and Figure S2.

A total of 159 patients (20.3%) in the pembrolizumab–chemotherapy group and 114 patients (29.2%) in the placebo–chemotherapy group had an event or died (hazard ratio, 0.65; 95% CI, 0.51 to 0.83) (Fig. 2A). The estimated 5-year event-free survival was 81.2% (95% CI, 78.3 to 83.8) in the pembrolizumab–chemotherapy group and 72.2% (95% CI, 67.4 to 76.4) in the placebo–chemotherapy group. The most common first event in the analysis of event-free survival in both groups was distant recurrence, with 77 events (9.8%) in the pembrolizumab–chemotherapy group and 56 events (14.4%) in the placebo–chemotherapy group (Table 1). Event-free survival in the prespecified subgroups is shown in Figure 2B and Figure S3.

The prespecified subgroup analysis of overall survival conducted according to the outcome (yes or no) of pathological complete response (ypT0–Tis ypN0) showed that in the population with a pathological complete response, 27 of 495 patients (5.5%) in the pembrolizumab–chemotherapy group and 17 of 217 patients (7.8%) in the placebo–chemotherapy group had died (absolute difference in 5-year survival, 0.7 percentage points; 95% CI, −2.9 to 4.3). In the population without a pathological complete response, 88 of 289 patients (30.4%) in the pembrolizumab–chemotherapy group and 68 of 173 patients (39.3%) in the placebo–chemotherapy group had died (absolute difference in 5-year survival, 6.1 percentage points; 95% CI, −2.7 to 14.9) (Fig. S4).

SAFETY

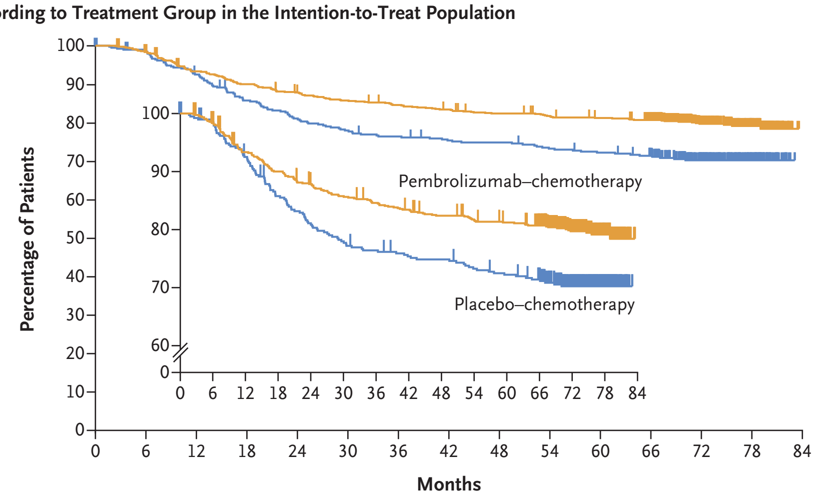
All the patients completed trial treatment by February 2020; the incidence of adverse events at the time of this analysis was similar to that reported previously.[10,11] Adverse events of grade 3 or higher that were considered by the investigator to be related to trial treatment occurred in 77.1% of 783 patients in the pembrolizumab–chemotherapy group and in 73.3% of 389 patients in the placebo–chemotherapy group, with neutropenia, decreased neutrophil count, and anemia being the most common (Table 2). Serious treatment-related adverse events occurred in 34.1% of the patients in the pembrolizumab–chemotherapy group and in 20.1% of those in the placebo–chemotherapy group. Treatment-related adverse events led to death in 4 patients (0.5%) and in 1 patient (0.3%), respectively.

Immune-mediated adverse events of any grade occurred in 35.0% of the patients in the pembrolizumab–chemotherapy group and in 13.1% of those in the placebo–chemotherapy group; grade 3 or higher events occurred in 13.0% and in 1.5% of the patients, respectively (Table 2). As reported previously,[10,11] a higher incidence of endocrine disorders was observed in the pembrolizumab–chemotherapy group than in the placebo–chemotherapy group. Immune-mediated adverse events led to death in two patients (0.3%) in the pembrolizumab–chemotherapy group and in no patients in the placebo–chemotherapy group.

DISCUSSION

In this phase 3 trial, neoadjuvant pembrolizumab combined with chemotherapy followed by adjuvant pembrolizumab resulted in a significant improvement, as compared with neoadjuvant chemotherapy alone, in overall survival among patients with previously untreated, high-risk, early-stage triple-negative breast cancer. After a median follow-up of 75.1 months, the 5-year overall survival was 4.9 percentage points higher with the addition of pembrolizumab. The benefit of pembrolizumab treatment with respect to overall sur-

A Event-free Survival According to Treatment Group in the Intention-to-Treat Population



No. at Risk

	0	6	12	18	24	30	36	42	48	54	60	66	72	78	84
Pembrolizumab–chemotherapy	784	769	728	702	681	665	654	644	633	625	618	602	409	164	0
Placebo–chemotherapy	390	382	358	330	312	300	293	287	285	278	273	264	178	76	0

B Subgroup Analyses of Event-free Survival

Subgroup	Pembrolizumab–Chemotherapy	Placebo–Chemotherapy	Hazard Ratio for Event or Death (95% CI)
	no. of patients with event/total no. (%)		
Overall	159/784 (20.3)	114/390 (29.2)	0.65 (0.51–0.83)
Nodal status			
Positive	102/408 (25.0)	65/196 (33.2)	0.72 (0.53–0.98)
Negative	57/376 (15.2)	49/194 (25.3)	0.56 (0.38–0.82)
Tumor size			
T1 to T2	91/580 (15.7)	76/290 (26.2)	0.55 (0.41–0.75)
T3 to T4	68/204 (33.3)	38/100 (38.0)	0.86 (0.58–1.28)
Carboplatin schedule			
Every 3 wk	65/334 (19.5)	45/167 (26.9)	0.68 (0.47–1.00)
Weekly	92/444 (20.7)	69/220 (31.4)	0.62 (0.46–0.85)
PD-L1 status			
CPS ≥1	128/656 (19.5)	86/317 (27.1)	0.68 (0.52–0.90)
CPS <1	31/128 (24.2)	28/69 (40.6)	0.53 (0.31–0.88)
Age			
<65 yr	130/700 (18.6)	98/342 (28.7)	0.61 (0.47–0.80)
≥65 yr	29/84 (34.5)	16/48 (33.3)	0.99 (0.54–1.82)

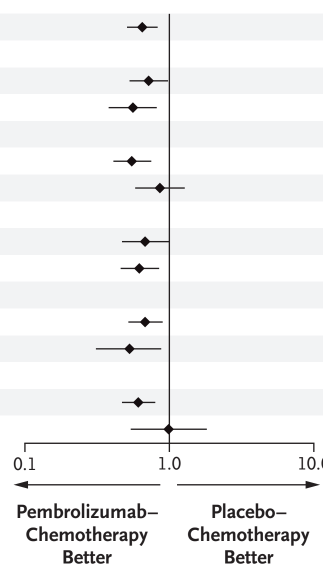


Figure 2. Event-free Survival.

Panel A shows Kaplan–Meier estimates of event-free survival according to treatment group in the intention-to-treat population. Tick marks represent data censored at the last time that the patient was known to be alive and without an event (defined as disease progression that precluded definitive surgery, local or distant recurrence, occurrence of a second primary tumor, or death from any cause). The inset shows the same data on an enlarged y axis. Panel B shows subgroup analyses of event-free survival. For the overall population and the PD-L1 subgroups, the analysis was based on the Cox regression model stratified according to nodal status (positive or negative), tumor size (T1 to T2 or T3 to T4), and frequency of carboplatin administration (once weekly or once every 3 weeks). For the other subgroups, the analysis was based on the unstratified Cox regression model.

Table 1. Summary of First Events in Analysis of Event-free Survival.

Event	Pembrolizumab–Chemotherapy (N = 784)	Placebo–Chemotherapy (N = 390)
	number (percent)	
Any event	159 (20.3)	114 (29.2)
Progression of disease that precluded definitive surgery	14 (1.8)	15 (3.8)
Local recurrence	33 (4.2)	20 (5.1)
Distant recurrence	77 (9.8)	56 (14.4)
Second primary cancer	16 (2.0)	10 (2.6)
Death	19 (2.4)	13 (3.3)

vival was seen in certain prespecified subgroups, including those defined according to PD-L1 expression, nodal status, and tumor size. The subgroup data are underpowered and require cautious interpretation.

Previous results from the KEYNOTE-522 trial showed a significant improvement in event-free survival with neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant pembrolizumab after a median follow-up of approximately 3 years.[11] In the present report, we describe updated results for event-free survival after a median follow-up of more than 6 years. At this time, a 35% relative reduction in the risk of disease progression that precluded definitive surgery, a local or distant recurrence, a second primary tumor, or death from any cause in the pembrolizumab–chemotherapy group, as compared with the placebo–chemotherapy group, was noted; these findings are consistent with the previous results.[11] The 5-year event-free survival was 9.0 percentage points higher with the addition of pembrolizumab.

The sustained benefit of pembrolizumab treatment with respect to event-free survival is notable given that recurrences in triple-negative breast cancer tend to occur earlier than in other breast cancer subtypes, with up to 90% occurring in the first 5 years.[5,25] Furthermore, the addition of pembrolizumab to chemotherapy resulted in a reduction in distant recurrences (9.8% in the pembrolizumab–chemotherapy group vs. 14.4% in the placebo–chemotherapy group), an established predictor of favorable long-term outcomes in line with the present results regarding overall survival. The benefit of pembrolizumab plus chemotherapy with respect to event-free survival was seen regardless of PD-L1 expression, nodal status, and tumor size. The consistency of these results with those reported previously[11] suggests a durable improvement in event-free survival with the addition of pembrolizumab in subgroups of patients known to have a poor prognosis.

At the first planned analysis of the KEYNOTE-522 trial, the addition of pembrolizumab to neoadjuvant chemotherapy significantly increased the percentage of patients who had a pathological complete response at the time of definitive surgery.[10] A numerically lower risk of death in the pembrolizumab–chemotherapy group than in the placebo–chemotherapy group was observed in the analysis of overall survival regardless of the outcome with respect to pathological complete response. Although the results of this nonrandomized comparison warrant cautious interpretation, they suggest that the overall survival benefit with pembrolizumab may exceed that expected from the improvement in pathological complete response alone.[26-31] This finding could be related to lower residual disease in the pembrolizumab–chemotherapy group at the end of the neoadjuvant phase, a change in the tumor biology as a result of the immunotherapy, or exposure to pembrolizumab in the adjuvant phase; however, the trial was not designed to distinguish the relative efficacy contributions in the long-term outcomes from the individual treatment phases.

The safety profile of pembrolizumab in this analysis was consistent with that in previous reports.[10,11] The adverse events were in line with the established safety profiles of pembrolizumab monotherapy and platinum-, taxane-, and anthracycline-containing chemotherapy, and the addition of pembrolizumab did not substantially exacerbate common chemotherapy-related toxic effects.[32,33] As expected with immunotherapy, immune-mediated adverse events occurred more frequently in the pembrolizumab–chemotherapy group, which were driven primarily by endocrinopathies. These events were generally low grade and successfully managed with treatment interruption, glucocorticoid administration, or hormone replacement. Nevertheless, some immune-mediated events may be irreversible and lead to long-term therapy, which highlights the need for early recognition and appropriate treatment, particularly for patients receiving potentially curative care. Data from studies of pembrolizumab in other

Table 2. Adverse Events in the Combined Neoadjuvant and Adjuvant Phases (As-Treated Population).*

Event	Pembrolizumab–Chemotherapy (N=783)		Placebo–Chemotherapy (N=389)	
	Any Grade	Grade ≥3	Any Grade	Grade ≥3
	number of patients (percent)			
Any adverse event	777 (99.2)	645 (82.4)	389 (100)	306 (78.7)
Treatment-related adverse event†	774 (98.9)	604 (77.1)	388 (99.7)	285 (73.3)
Nausea	495 (63.2)	27 (3.4)	245 (63.0)	6 (1.5)
Alopecia	471 (60.2)	0	220 (56.6)	0
Anemia	429 (54.8)	141 (18.0)	215 (55.3)	58 (14.9)
Neutropenia	367 (46.9)	270 (34.5)	185 (47.6)	130 (33.4)
Fatigue	330 (42.1)	28 (3.6)	151 (38.8)	7 (1.8)
Diarrhea	238 (30.4)	20 (2.6)	98 (25.2)	5 (1.3)
Elevated alanine aminotransferase level	204 (26.1)	43 (5.5)	98 (25.2)	9 (2.3)
Vomiting	200 (25.5)	19 (2.4)	86 (22.1)	6 (1.5)
Asthenia	198 (25.3)	28 (3.6)	102 (26.2)	9 (2.3)
Rash	196 (25.0)	12 (1.5)	66 (17.0)	1 (0.3)
Constipation	188 (24.0)	0	85 (21.9)	0
Decreased neutrophil count	185 (23.6)	146 (18.6)	112 (28.8)	90 (23.1)
Elevated aspartate aminotransferase level	157 (20.1)	20 (2.6)	63 (16.2)	1 (0.3)
Peripheral neuropathy	154 (19.7)	15 (1.9)	84 (21.6)	4 (1.0)
Immune-mediated adverse event‡	274 (35.0)	102 (13.0)	51 (13.1)	6 (1.5)
Hypothyroidism	118 (15.1)	4 (0.5)	22 (5.7)	0
Severe skin reactions	45 (5.7)	37 (4.7)	4 (1.0)	1 (0.3)
Hyperthyroidism	41 (5.2)	2 (0.3)	7 (1.8)	0
Gastritis	27 (3.4)	2 (0.3)	8 (2.1)	1 (0.3)
Adrenal insufficiency	20 (2.6)	8 (1.0)	0	0
Pneumonitis	17 (2.2)	7 (0.9)	6 (1.5)	2 (0.5)
Thyroiditis	16 (2.0)	2 (0.3)	5 (1.3)	0
Hypophysitis	15 (1.9)	10 (1.3)	1 (0.3)	0

* Listed are adverse events that occurred during the treatment period or within 30 days after the treatment period (or, for serious adverse events, within 90 days after the treatment period) in descending order of frequency in the pembrolizumab–chemotherapy group. The as-treated population included all the patients who had undergone randomization and received at least one trial drug, underwent surgery, or both. The severity of adverse events was graded according to the Common Terminology Criteria for Adverse Events, version 4.0, of the National Cancer Institute.

† Treatment-related adverse events were events that were attributed to a trial treatment by the investigators. Treatment-related adverse events that occurred in at least 20% of the patients in either treatment group are reported. Patients may have had more than one event. Grade 5 treatment-related adverse events were sepsis and multiple organ dysfunction syndrome (in one patient) and pneumonitis, pulmonary embolism, and autoimmune encephalitis (in one patient each) in the pembrolizumab–chemotherapy group and septic shock (in one patient) in the placebo–chemotherapy group.

‡ Immune-mediated adverse events were determined according to a list of terms specified by the sponsor, regardless of attribution to any trial treatment by the investigators. Immune-mediated adverse events that occurred in at least 15 patients in either treatment group are reported. Grade 5 immune-mediated adverse events were pulmonary embolism and autoimmune encephalitis (in 1 patient each) in the pembrolizumab–chemotherapy group.

cancer types have shown no signal for late toxic effects,[34,35] which provides further evidence of the long-term safety profile. Still, continued monitoring is needed, because delayed immune-mediated toxic effects in a minority of patients have been reported.[36]

KEYNOTE-522 is a phase 3, prospective, randomized, placebo-controlled trial of neoadjuvant and adjuvant pembrolizumab treatment showing significant improvements in overall survival, event-free survival, and pathological complete response with combined immunotherapy and standard chemotherapy in breast cancer. A key strength of the trial is the inclusion of a control group of patients who received standard-care platinum-, taxane-, and anthracycline-containing chemotherapy, which allowed comparison of the pembrolizumab–chemotherapy regimen with the most effective neoadjuvant chemotherapy in early-stage triple-negative breast cancer. The present analyses also had some limitations. The trial was not designed to discern the relative efficacy contributions of the neoadjuvant and adjuvant treatment phases. Furthermore, adjuvant capecitabine was not included in the treatment protocol because the results of the Capecitabine for Residual Cancer as Adjuvant Therapy trial showing a survival benefit in patients with triple-negative breast cancer[37] were reported after the trial was designed. In addition, analyses in patient subgroups were exploratory, and the results in small subgroups could be due to random variation and warrant cautious interpretation. Analyses of molecular biomarkers that might predict clinical response to pembrolizumab are ongoing.

In this trial, neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant pembrolizumab significantly improved overall survival as compared with neoadjuvant chemotherapy alone. The benefit with respect to overall survival was observed in certain patient subgroups defined according to prognostic risk factors, including lymph-node involvement and tumor size, and was observed regardless of PD-L1 expression status and the outcome with respect to pathological complete response. Pembrolizumab–chemotherapy continued to be associated with improved event-free survival after a median follow-up of more than 6 years. These results provide further support for pembrolizumab plus neoadjuvant chemotherapy followed by adjuvant pembrolizumab as treatment for high-risk, early-stage triple-negative breast cancer.

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A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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Supplementary Appendix

Supplement to: Schmid P, Cortes J, Dent R, et al. Overall survival with pembrolizumab in early-stage triple-negative breast cancer. N Engl J Med 2024;391:1981-91. DOI: 10.1056/NEJMoa2409932

This appendix has been provided by the authors to give readers additional information about the work.

Supplementary Appendix

For

Overall Survival with Pembrolizumab in Early-Stage Triple-Negative Breast Cancer

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United States	Henry Ford Hospital	Ali, Haythem
	Northwest Cancer Specialists, P.C.	Andersen, Jay C.
	New England Cancer Specialists	Battelli, Chiara
	Magee Women's Hospital	Brufsky, Adam//Emens, Leisha Ann/Jankowitz, Rachel
	Texas Oncology-Memorial City	Cairo, Michelina/Holmes, Frankie Ann
	Houston Methodist Cancer Center	Chang, Jenny
	Medical Oncology Associates (Summit Cancer Centers)	Chaudhry, Arvind

	Peninsula Cancer Institute, LLC	Chevinsky, Aaron/MacLaughlin, William
	North Shore Oncology Hematology Associates	Chung, Elizabeth K./Weyburn, Thomas D.
	Virginia Oncology Associates	Danso, Michael A.
	University of Colorado Cancer Center	Diamond, Jennifer Robinson
	University of Virginia Comprehensive Cancer Center	Dillon, Patrick
	Rhode Island Hospital	Fenton, Mary Anne/Lopresti, Mary
	Scottsdale Healthcare Hospitals D/B/A HonorHealth	Gordon, Michael S./Sachdev, Jasgit
	Moncrief Cancer Institute	Haley, Barbara B.
	Parkland Health and Hospital System	Haley, Barbara B.
	Simmons Cancer Center	Haley, Barbara B.
	UT Southwestern Medical Center	Haley, Barbara B.
	Moncrief Cancer Institute	Haley, Barbara B.
	Broome Oncology, LLC	Harris, Ronald P.
	Texas Oncology-San Antonio Northeast	Harroff, Allyson/Wilks, Sharon T.
	Bon Secours St. Francis Medical Center-Oncology Research	Irvin, William J.
	Pacific Cancer Care	Koontz, Michael Zachary/Stampleman, Laura
	Goshen Center for Center Care	Kio, Ebenezer/Bruetman, Daniel G.
	Minnesota Oncology Hematology, P.A.	Larson, Timothy G./Tsai, Michaela
	University of Iowa Hospital and Clinics	Leone, Jose/Phadke, Sneha
	YVMH dba Virginia Mason Memorial/North Star Lodge Cancer Center	Mannem, Siva/Obulareddy, Sri/Jones, Vicky E.
	TOI Clinical Research	Marathe, Omkar/Thara, Eddie
	TriHealth Cancer Institute-Good Samaritan Hospital	Mehta, Apurva
	Helen F. Graham Cancer Center & Research Institute	Misleh, Jamal/Guarino, Michael J.
	Cedars Sinai Medical Center	Mita, Monica M./Basho, Reva/McArthur, Heather
	University of Chicago Medical Center	Nanda, Rita
	Orchard Healthcare Research Inc.	Oliff, Ira A.
	Texas Oncology - Baylor Charles A. Sammons Cancer Center	OShaughnessy, Joyce A.
	Providence Portland Medical Center	Page, David
	Texas Oncology-Austin Central	Patt, Debra A.
	Yale University School of Medicine	Pusztai, Lajos
	Texas Oncology-Northeast Texas	Richards, Donald A.
	Nyack Hospital Infusion Center	Rybalova, Irina

	NorthShore University Health System - Evanston Hospital	Saha, Poornima
	Seattle Cancer Care Alliance	Specht, Jennifer
	Virginia Cancer Specialists, PC	Spira, Alexander I./Denduluri, Neelima
	Texas Oncology - Plano East	Stokoe, Christopher T.F.
	Arizona Oncology Associates P.C.- HOPE	Swart, Rachel
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	The West Clinic, P.C.	Vidal, Gregory
	Oncology/Hematology Care Clinical Trials, LLC	Ward, Patrick
	Kadlec Clinic Hematology and Oncology	Zhuo, Ying

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Supplemental Methods.

Prespecified subgroups

To evaluate the consistency of the treatment effects across various subgroups, the estimate of the between-group treatment effect (with a nominal 95% CI) for the primary endpoints was estimated and plotted within each category of the following classification variables:

- Nodal status: Positive vs. Negative
- Tumor size: T1/T2 vs. T3/T4
- Choice of Carboplatin: Q3W vs. Weekly
- Tumor PD-L1 status
- Menopausal status: pre-menopausal vs. post-menopausal
- Age: <65 years vs. ≥65 years
- Geographic region: Europe/Israel/North America/Australia vs. Asia vs. Rest of World
- Ethnic origin: Hispanic vs. Non-Hispanic
- ECOG performance status: 0 vs. 1
- HER2 status: IHC 2+ (but FISH-) vs. IHC 0-1+
- LDH: >upper limit of normal (ULN) vs. ≤ULN

Supplemental Results

Overall survival

The proportional hazards assumption does not hold for overall survival at this interim analysis. The hazard ratio for overall survival based on the protocol-specified Cox regression model is 0.66 (95% CI, 0.50-0.87).

Figure S1. Trial Enrollment and Outcomes of the Patients at the Seventh Interim Analysis. Shown are data until the data-cutoff date of March 22, 2024. Patients did not have to complete all neoadjuvant therapy to undergo surgery. No patients remained on treatment at the seventh interim analysis. ITT, intention-to-treat.

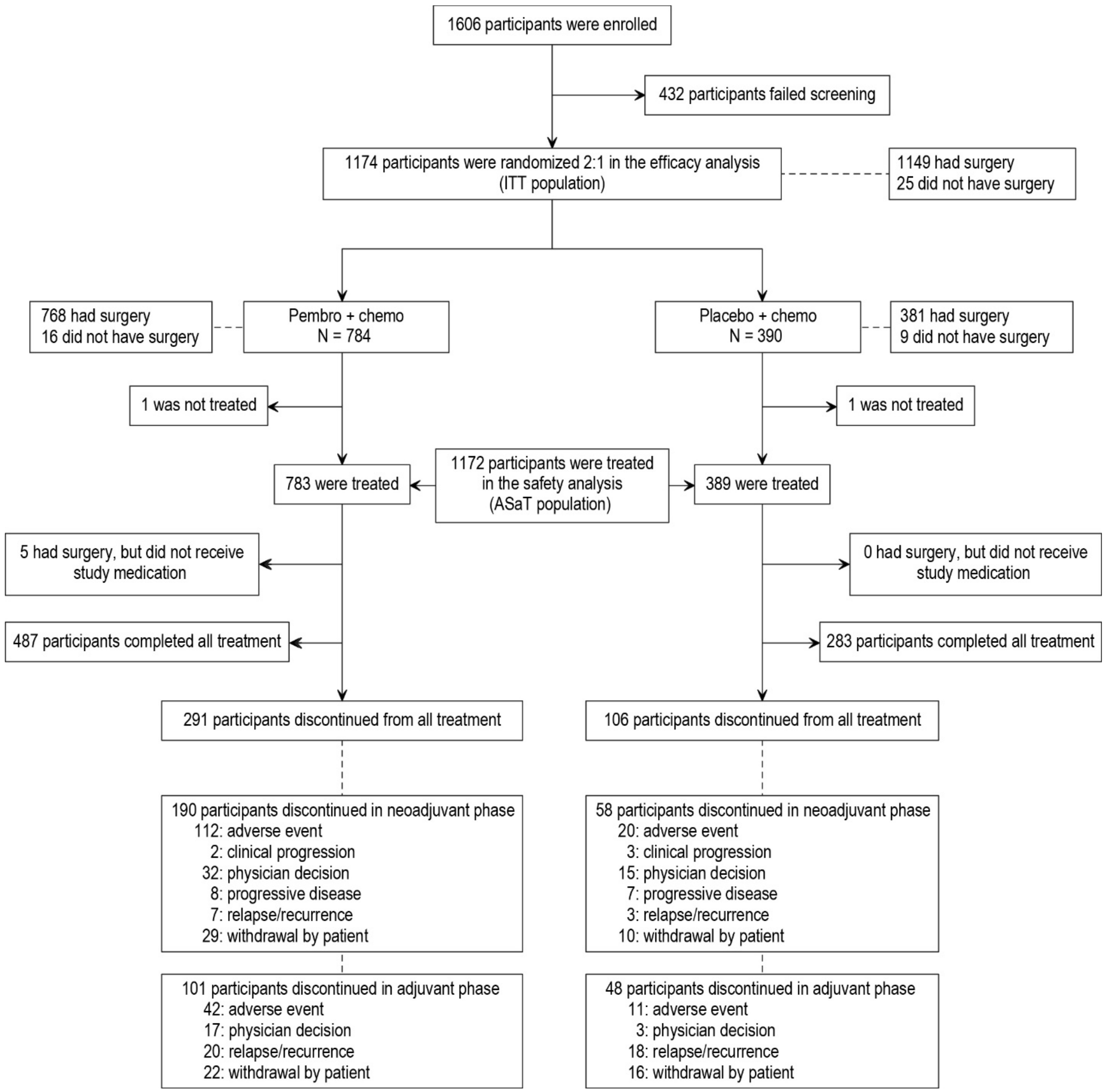

Fig. S2. Additional Subgroup Analyses of Overall Survival. Kaplan-Meier estimates of 5-year overall survival in additional prespecified subgroups is shown. The difference in 5-year overall survival and 95% confidence intervals for the difference are reported, assuming that the Kaplan-Meier survival rates at 5 years in both treatment groups follow the asymptomatic normal distribution independently. Eastern Cooperative Oncology Group (ECOG) performance-status (PS) scores range from 0 to 5, with higher scores indicating greater disability. Patients with missing values in a subgroup category variable are not included in the subgroup analysis.

Subgroup	Pembrolizumab-Chemotherapy	Placebo-Chemotherapy	Difference in 5-Year Overall Survival
	no. of patients with event/total no. (%)		*percentage points (95% CI)*
Overall	115/784 (14.7)	85/390 (21.8)	4.9 (0.3 to 9.4)
Menopausal status			
Pre-menopausal	55/438 (12.6)	39/221 (17.6)	4.0 (-1.7 to 9.7)
Post-menopausal	60/345 (17.4)	46/169 (27.2)	6.0 (-1.3 to 13.3)
Geographic region			
EU/Israel/NA/Australia	90/607 (14.8)	60/285 (21.1)	3.2 (-1.9 to 8.3)
Asia	12/136 (8.8)	16/80 (20.0)	10.8 (1.0 to 20.6)
Rest of World	13/41 (31.7)	9/25 (36.0)	4.3 (-19.3 to 27.9)
Ethnic origin			
Hispanic	23/85 (27.1)	14/39 (35.9)	2.5 (-14.5 to 19.5)
non-Hispanic	79/616 (12.8)	59/307 (19.2)	4.0 (-0.8 to 8.8)
HER2 status			
2+ by IHC (but FISH-)	28/188 (14.9)	22/104 (21.2)	3.4 (-5.4 to 12.2)
0-1+ by IHC	87/595 (14.6)	63/286 (22.0)	5.3 (0.0 to 10.6)
LDH			
>ULN	26/149 (17.4)	24/80 (30.0)	7.6 (-3.5 to 18.7)
≤ULN	88/631 (13.9)	60/309 (19.4)	3.9 (-1.0 to 8.8)
ECOG PS			
0	92/678 (13.6)	74/341 (21.7)	6.6 (1.7 to 11.4)
1	23/106 (21.7)	11/49 (22.4)	-6.5 (-18.9 to 6.0)

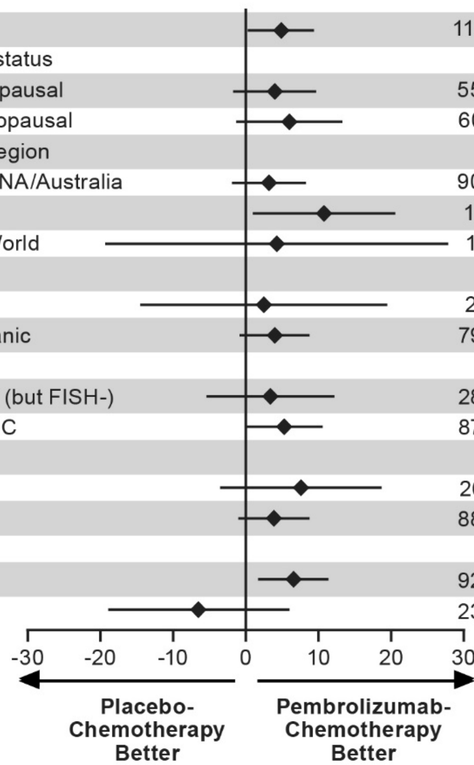

Fig. S3. Additional Subgroup Analyses of Event-Free Survival. An analysis of event-free survival in additional prespecified subgroups is shown. For the overall population, the analysis was based on the Cox regression model stratified according to nodal status (positive or negative), tumor size (T1 to T2 or T3 to T4), and frequency of carboplatin administration (once weekly or once every 3 weeks). For the other subgroups, the analysis was based on the unstratified Cox regression model. Eastern Cooperative Oncology Group (ECOG) performance-status (PS) scores range from 0 to 5, with higher scores indicating greater disability. Patients with missing values in a subgroup category variable are not included in the subgroup analysis. The proportional hazards assumption did not hold for the ECOG=1 subgroup; the 5-year rates of event-free survival were 75.2% in the pembrolizumab–chemotherapy group and 69.4% in the placebo–chemotherapy group (difference [95% CI] 5.8% [-9.5%, 21.2%]).

Subgroup	Pembrolizumab-Chemotherapy (no. of patients with event/total no. (%))	Placebo-Chemotherapy (no. of patients with event/total no. (%))	Hazard Ratio (95% CI)
Overall	159/784 (20.3)	114/390 (29.2)	0.65 (0.51 to 0.83)
Menopausal status			
Pre-menopausal	75/438 (17.1)	56/221 (25.3)	0.64 (0.46 to 0.91)
Post-menopausal	84/345 (24.3)	58/169 (34.3)	0.66 (0.48 to 0.93)
Geographic region			
EU/Israel/NA/Australia	126/607 (20.8)	83/285 (29.1)	0.68 (0.52 to 0.90)
Asia	18/136 (13.2)	22/80 (27.5)	0.43 (0.23 to 0.81)
Rest of World	15/41 (36.6)	9/25 (36.0)	0.95 (0.42 to 2.18)
Ethnic origin			
Hispanic	29/85 (34.1)	15/39 (38.5)	0.80 (0.43 to 1.49)
non-Hispanic	110/616 (17.9)	83/307 (27.0)	0.63 (0.47 to 0.83)
HER2 status			
2+ by IHC (but FISH-)	38/188 (20.2)	29/104 (27.9)	0.70 (0.43 to 1.14)
0-1+ by IHC	121/595 (20.3)	85/286 (29.7)	0.64 (0.49 to 0.85)
LDH			
>ULN	33/149 (22.1)	31/80 (38.8)	0.54 (0.33 to 0.88)
≤ULN	125/631 (19.8)	82/309 (26.5)	0.71 (0.53 to 0.93)
ECOG PS			
0	132/678 (19.5)	98/341 (28.7)	0.63 (0.49 to 0.82)
1	27/106 (25.5)	16/49 (32.7)	0.80 (0.43 to 1.49)

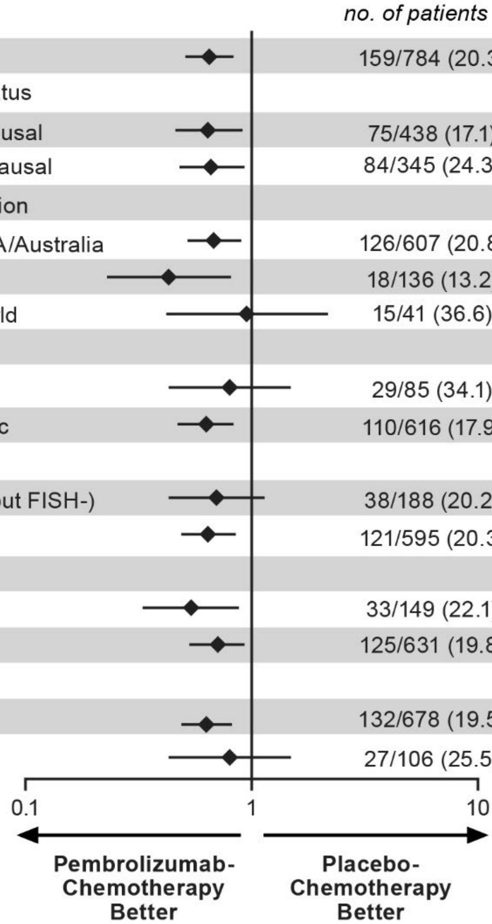

Fig. S4. Kaplan-Meier Estimates of Overall Survival by Pathological Complete Response (ypT0/Tis ypN0) According to Treatment Group in the Intention-to-Treat Population. Tick marks represent data censored at the last time the patient was known to be alive.

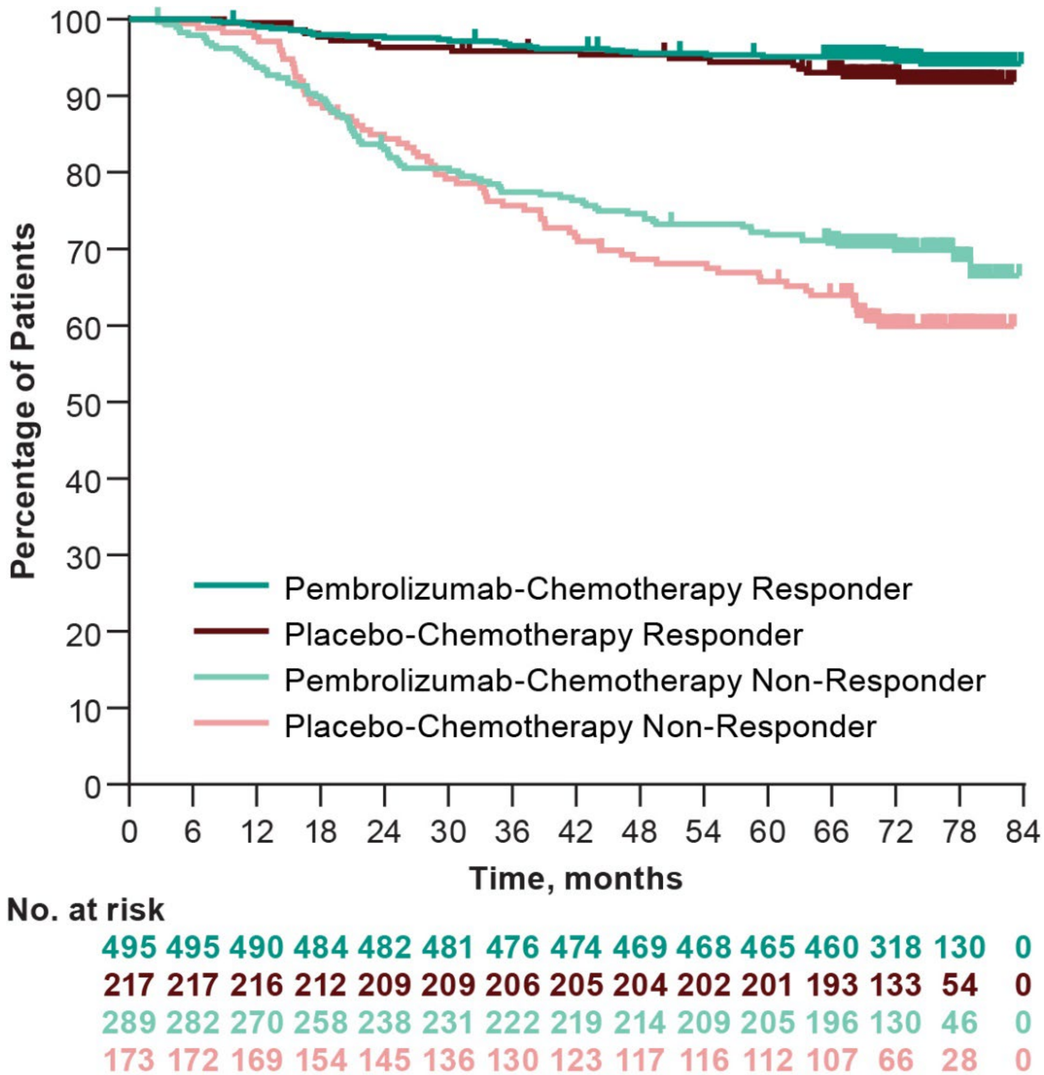

Table S1. Demographic and Disease Characteristics at Baseline.*

Characteristic	Pembrolizumab–Chemotherapy (N=784)	Placebo–Chemotherapy (N=390)
Age		
Median (range) — yr	49 (22-80)	48 (24-79)
<65 yr — no. (%)	700 (89.3)	342 (87.7)
Female sex — no. (%)	783 (99.9)	390 (100.0)
Geographic region — no. (%)		
Europe	388 (49.5)	180 (46.2)
Asia	166 (21.2)	91 (23.3)
North America	166 (21.2)	78 (20.0)
Australia	23 (2.9)	16 (4.1)
Rest of world	41 (5.2)	25 (6.4)
Race		
White	504 (64.3)	242 (62.1)
Asian	149 (19.0)	89 (22.8)
Missing†	65 (8.3)	31 (7.9)
Black or African American	38 (4.8)	15 (3.8)
American Indian or Alaska Native	14 (1.8)	7 (1.8)
Multiple	13 (1.7)	6 (1.5)
Native Hawaiian or Other Pacific Islander	1 (0.1)	0
Ethnicity		
Not Hispanic or Latino	616 (78.6)	307 (78.7)
Hispanic or Latino	85 (10.8)	39 (10.0)
Not Reported	46 (5.9)	28 (7.2)
Unknown	19 (2.4)	11 (2.8)
Missing†	18 (2.3)	5 (1.3)
Menopausal status — no. (%)		
Pre-menopausal	438 (55.9)	221 (56.7)
Post-menopausal	345 (44.0)	169 (43.3)
Missing‡	1 (0.1)	0 (0.0)
PD-L1 status§ — no. (%)		
CPS ≥1	656 (83.7)	317 (81.3)
CPS <1	128 (16.3)	69 (17.7)
Missing	0 (0.0)	4 (1.0)
ECOG performance status		
0	678 (86.5)	341 (87.4)
1	106 (13.5)	49 (12.6)
Lactase dehydrogenase — no. (%)		
≤ULN	631 (80.5)	309 (79.2)
>ULN	149 (19.0)	80 (20.5)
Missing	4 (0.5)	1 (0.3)
Choice of carboplatin — no. (%)		
Every 3 weeks	335 (42.7)	167 (42.8)
Weekly	449 (57.3)	223 (57.2)

Primary tumor classification— no. (%)		
T1/T2	580 (74.0)	290 (74.4)
T3/T4	204 (26.0)	100 (25.6)
Nodal involvement — no. (%)		
Positive	405 (51.7)	200 (51.3)
Negative	379 (48.3)	190 (48.7)
Overall disease stage — no. (%)		
I	0 (0.0)	1 (0.3)
II	590 (75.3)	291 (74.6)
III	194 (24.7)	98 (25.1)
HER2 status — no. (%)		
0-1 by IHC	595 (75.9)	286 (73.3)
2+ by IHC[‖]	188 (24.0)	104 (26.7)
Missing	1 (0.1)	0 (0.0)

*Based on data from the intention-to-treat population. Percentages may not total 100 because of rounding. PD-L1 denotes programmed death ligand 1; ECOG denotes Eastern Cooperative Oncology Group; HER2 denotes human epidermal growth factor receptor 2; IHC denotes immunohistochemistry.
[†]Information on race and ethnicity of trial participants is based on self-report and unavailable in some countries due to local laws and practices.
[‡]From one male patient.
[§]The PD-L1 combined positive score was defined as number of PD-L1–positive cells (tumor cells, lymphocytes, and macrophages) divided by total number of tumor cells × 100.
[‖]All tumors with HER2 expression 2+ by IHC were negative for HER2 amplification by in situ hybridization.

Table S2. Representativeness of Patient Population.

Disease under investigation	Triple-negative breast cancer (TNBC).
Special considerations related to	
Sex and gender	Breast cancer is the most diagnosed cancer among women, and the leading cause of cancer deaths globally[1], with TNBC accounting for 15% to 20% of all breast cancer subtypes.[2] TNBC in men is extremely rare.
Age	Compared with other forms of breast cancer, rates of TNBC are higher in young persons, with many patients being diagnosed before the age of 50 years.[3,4]
Race or ethnic group	TNBC affects Black persons disproportionately in the United States.[5] Several studies suggest a similar rate of among Hispanic and non-Hispanic White persons.[6] Compared with non-Hispanic White persons, Asian persons have significantly lower rates of premenopausal TNBC, but rates do not decline with increasing age.[7]
Geography	In 2022, female breast cancer was the second leading cause of global cancer incidence, with approximately 2.3 million new cases, comprising 11.6% of all cancer cases.[1] Of the global breast cancer burden, it has been estimated that ~170 000 are TNBC.[8] Significant heterogeneity was reported among different groups of Black persons by birthplace, with the highest prevalence among US-born and Western-African–born Black women, followed by Caribbean-born, and Eastern-African-born Black women.[9]
Other considerations	The incidence of breast cancer and mortality rates vary greatly within and between countries. Ethnic predisposition, family history, genetic associations, and modifiable risk factors are inextricably intertwined.
Overall representativeness of this trial	The patients in this trial demonstrated the expected sex and age distribution. The proportion of Black persons who underwent randomization was small (4.5%). Among the patients enrolled in North America, 15.2% were Black, a relatively larger proportion than the total population distribution of Black persons in the United States. Most patients were enrolled in Europe (48.4%) and North America (20.8%); no patient was enrolled in Africa. The information on race and ethnicity in this study is based on self-report and unavailable in some countries due to local laws and practices.

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Table S3. Summary of Drug Exposure in the Combined Phases.

Event	Pembrolizumab–Chemotherapy (N=783)	Placebo–Chemotherapy (N=389)
All drugs		
n	778	389
Median weeks (range)	57.9 (0.1 – 95.3)	59.1 (0.1 – 86.1)
Pembrolizumab (200 mg Q3W)		
n	778	0
Median weeks (range)	57.9 (0.1 – 95.3)	0
Median no. administrations (range)	17.0 (1.0. – 17.0)	0
Placebo (Q3W)		
n	0	389
Median weeks (range)	0	59.1 (0.1 – 86.1)
Median no. administrations (range)	0	17.0 (1.0 – 17.0)
Carboplatin (weekly)		
n	444	220
Median weeks (range)	11.1 (2.0 – 26.1)	11.1 (2.3 – 17.1)
Median no administrations (range)	12.0 (3.0 – 12.0)	12.0 (3.0 – 12.0)
Carboplatin (Q3W)		
n	334	167
Median weeks (range)	9.6 (0.1 – 16.4)	9.4 (3.0 – 15.1)
Median no administrations (range)	4.0 (1.0 – 4.0)	4.0 (2.0 – 4.0)
Paclitaxel (weekly)		
n	778	389
Median weeks (range)	11.3 (0.1 – 26.1)	11.3 (0.1 – 17.1)
Median no. administrations (range)	12.0 (1.0 – 13.0)	12.0 (1.0 – 12.0)
Doxorubicin (Q3W)		
n	488	247
Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 13.3)
Median no. administrations (range)	4.0 (1.0 – 4.0)	4.0 (1.0 – 4.0)
Epirubicin (Q3W)		
n	238	122

Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 12.0)
Median no administrations (range)	4.0 (1.0 – 5.0)	4.0 (1.0 – 4.0)
Cyclophosphamide (Q3W)		
n	726	369
Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 13.3)
Median no administrations (range)	4.0 (1.0 – 4.0)	4.0 (1.0 – 4.0)